6CS012 - Assessment Submission

Islington College,Nepal

Document Details

Submission ID

trn:oid:::3618:138370701

Submission Date

May 10, 2026, 4:15 PM GMT+5:45

Download Date

May 14, 2026, 12:12 PM GMT+5:45

File Name

2431154_PranishaMaharjan_Report_Image_Classification.pdf

File Size

275.6 KB

8 Pages

1,421 Words

10,200 Characters

14% Overall Similarity

The combined total of all matches, including overlapping sources, for each database.

Filtered from the Report

- Bibliography
- Quoted Text
- Cited Text
- Small Matches (less than 12 words)

Exclusions

- 2 Excluded Sources
- 1 Excluded Match

Match Groups

12 Not Cited or Quoted 14%
Matches with neither in-text citation nor quotation marks

0 Missing Quotations 0%
Matches that are still very similar to source material

0 Missing Citation 0%
Matches that have quotation marks, but no in-text citation

0 Cited and Quoted 0%
Matches with in-text citation present, but no quotation marks

Top Sources

3%	Internet sources
0%	Publications
13%	Submitted works (Student Papers)

Match Groups

- **12** Not Cited or Quoted 14%
  Matches with neither in-text citation nor quotation marks
- **0** Missing Quotations 0%
  Matches that are still very similar to source material
- **0** Missing Citation 0%
  Matches that have quotation marks, but no in-text citation
- **0** Cited and Quoted 0%
  Matches with in-text citation present, but no quotation marks

Top Sources

3% Internet sources
0% Publications
13% Submitted works (Student Papers)

Top Sources

The sources with the highest number of matches within the submission. Overlapping sources will not be displayed.

1	Submitted works	Islington College,Nepal on 2026-05-10	4%
2	Internet	arxiv.org	2%
3	Submitted works	Islington College,Nepal on 2026-05-09	1%
4	Submitted works	Islington College,Nepal on 2026-05-10	1%
5	Submitted works	Islington College,Nepal on 2026-05-09	<1%
6	Submitted works	Islington College,Nepal on 2026-05-10	<1%
7	Submitted works	Islington College,Nepal on 2026-05-10	<1%
8	Internet	ijarcsms.com	<1%
9	Submitted works	Islington College,Nepal on 2026-05-10	<1%
10	Submitted works	Islington College,Nepal on 2026-05-10	<1%

UNIVERSITY PARTNER

UNIVERSITY OF WOLVERHAMPTON

HERALD COLLEGE KATHMANDU

(6CS012)

Artificial Intelligence & Machine Learning

Part – 2

Sign Language Detection

Student Id	: [2431154]
Student Name	: [Pranisha Maharjan]
Group	: (L6CG11)
Tutor	: [Bijay Ghimire]
Module Leader	: [Siman Giri]
Submitted on	: 2026-05-19

Abstract

This project focuses on deep learning models to classify sign language images into multiple gesture categories. The baseline architecture involved a CNN with three convolutional layers and a deeper CNN with additional convolutional layers and used different methods of regularization. The dataset included 10,792 sign language images and the images were resized, normalized, and augmented as part of pre-processing steps. Among the three models, the complex CNN model without Dropout gave a higher validation accuracy score of 70.76%, better than the complex CNN with Dropout, giving an accuracy score of 63.25%, and the simple CNN, whose accuracy score was 61.57%. The mobile-net v2 transfer learning model performed the best, with an accuracy score of about 89.04% which shows the benefits of using pre-trained models for feature extraction. Moreover, transfer learning models provide the best accuracy scores compared to training from scratch, as they leverage pre-trained models to learn features. The Adam optimizer converged faster and more stably compared to SGD. Overfitting, underfitting, and dataset challenges were the main issues in the current experiment. However, Batch Normalization and regularization techniques helped overcome these issues.

Table Of Contents

1. Introduction

Image Classification is one of the most significant applications of deep learning and computer vision. This process refers to automatic identification and classification of objects within an image. The deep learning algorithms, particularly CNNs, have been extremely effective for this task because they are able to automatically extract useful visual features from the raw images.

The current research project will explore sign language image classification. Recognition of sign languages may contribute to improving communication among people with hearing disabilities and the wider population. Sign language recognition models can be employed in educational purposes, assistive technologies, and human-machine interaction. Moreover, deeper CNN models and transfer learning models increase efficiency through learning complicated features from images.

2. Dataset

The dataset used in the project is Sign Language Detection which was shared by tutor in google drive and accessed through Google Colab. The dataset consists of 10,792 total images and 38 classes. The dataset contained multi-class image classification dataset and images and RGB images with different resolutions and qualities, but images were resized before training. The dataset was split into 80% training images (8648) and 20% into validation images (2144).

2.1 Preprocessing

Different types of preprocessing techniques are applied before training:

- Resizing : Images were resized to 64 × 64 pixels for CNN models and 224 × 224 pixels for MobileNetV2.
- Augmentation : Training data are augmented using random rotations, zooming, width shifts, height shifts, and horizontal flipping to improve model generalization.
- Normalization : Pixel values were scaled to the range [0, 1] by dividing by 255.
- Cleaning : Corrupted images were removed using PIL.

2.2 Data Collection and Challenges

The challenges faced with dataset preparation were corrupt image files, imbalance in classes, changes in lighting and hand position, and variations in image quality.

1

3. Methodology

3.1 Baseline Model

The baseline CNN model was designed using the Keras Sequential API. The architecture includes:

2

- 3 Convolutional Layers with 32, 64, and 128 filters using 3 × 3 kernels and ReLU activation.
- 3 MaxPooling Layers with 2 × 2pooling to reduce spatial dimensions.

10

- 3 Fully Connected (Dense) Layers with 256, 128, and 64 neurons.
- Output Layer with SoftMax activation for multi-class classification.
- Input Shape: 64 × 64 × 3RGB images.

5

3.2 Deeper Model

The deeper CNN model was developed to improve feature extraction and model generalization.

- Additional Convolutional Layers with increased filter sizes (32, 64, 128, 256, and 512).
- Batch Normalization Layers after convolution operations to stabilize training.
- MaxPooling Layers for dimensionality reduction.
- Dropout Layers with dropout rates of 0.3 - 0.5 to reduce overfitting.
- Dense Layers with higher learning capacity.
- Output Layer using SoftMax activation.

4

3.3 Loss Function and Optimizer

Loss Function : The models used Categorical Cross-Entropy Loss for multi-class image classification.

Optimizer :

- Adam : Faster convergence with adaptive learning rates and stable training performance, achieved higher validation accuracy compared to SGD.
- SGD : Used learning rate 0.01 and momentum 0.9 with slower convergence ,training was less stable and achieved lower validation accuracy.

3.4 Hyperparameters

The following hyperparameters were used during training:

- Batch Size: 32
- Epochs: 25–30 depending on the model.

- Callbacks Used : The hyperparameters used during training are: EarlyStopping, ModelCheckpoint , ReduceLROnPlateau.

4. Experiments and Results

4.1 Baseline Vs Deeper Architecture

Baseline CNN :

Baseline CNN gave around 61.57% validation accuracy. The network was able to learn the basic features like hand shapes, edges, and gestures. Yet, the difference in performance between training and validation indicated some degree of overfitting. The shallow architecture could not learn complicated patterns from the input.

Deeper CNN :

Deep CNN resulted in about 63.25% validation accuracy. Using extra convolution layers enabled the network to learn more sophisticated features from images of sign language. Batch Normalization added more stability to training, whereas Dropout prevented overfitting.

4.2 Computational Efficiency

- Baseline CNN: 4–5 min/epoch, 25 epochs, lower overall training time.
- Deeper CNN: 7–9 min/epoch, 30 epochs, higher computational cost due to additional layers.
- MobileNetV2**:** 8–12 min/epoch , 20 epochs ,during feature extraction and fine-tuning.
- Hardware: Training was performed using Google Colab with GPU acceleration, which significantly reduced execution time.

Analysis: The baseline CNN was the most computationally efficient model, while the deeper CNN and MobileNetV2 required more GPU memory and longer training time because of increased architectural complexity.

4.3 Training with Different Optimizers

4.4 Challenges in Training

Overfitting : The baseline CNN showed clear overfitting, where training accuracy was high, but validation performance dropped. This was improved in the deeper CNN by using Batch Normalization and Dropout layers.

Class Imbalance : The dataset was not fully balanced, and the Normal class performed poorly in terms of precision and recall compared to other classes.

Underfitting : When the model was trained using SGD optimizer (learning rate 0.01, momentum 0.9), it learned slowly and gave lower accuracy compared to Adam.

Computational Cost : Training multiple models (Baseline CNN, Deeper CNN, and Transfer Learning models) required significant GPU time, with a total training duration of around 4.5–5 hours in Google Colab.

Data Handling**:** No corrupted images were found during preprocessing, as all images were successfully cleaned before training

5. Fine-tuning or Transfer Learning

The pre-trained MobileNetV2 architecture was fine-tuned:

Model Configuration:

The classifier included: GlobalAveragePooling2D, Dense (256, ReLU activation function, Dropout rate = 0.5), Dense (128, ReLU), and Dense (38, softmax).

Input shape: 224 x 224 x 3 (RGB).

Feature Extraction:

Validation accuracy: 87%.

Remarks: Good stability during training and efficient feature extraction using pre-trained ImageNet features.

Fine-Tuning:

The top layers were unfrozen and trained.

Validation accuracy: 89.04%.

Remarks: Improvement in classification efficiency and generalization through fine-tuning.

Evaluation:

The highest validation accuracy recorded: 89.04%.

Remarks: MobileNetV2 outperformed all other CNN architecture in terms of efficiency.

5.1 Comparison with Scratch Models

Model	Accuarcy	Observations
Baseline CNN	61.57%	Simple architecture with fast training, but limited feature extraction
Deeper CNN	63.25%	Increased learning with depth and regularizes
Deeper CNN without Dropout	70.76%	Improvement in Accuracy due to Removal of Dropout
ResNet50	89.04%	Highest accuracy, as it used pre-trained features extractor

6. Conclusion and Future Work

The project was focused on comparing different types of CNNs designed for sign language image recognition tasks. Among the created custom CNNs, the deeper network without Dropout outperformed the rest with 70.76% validation accuracy, whereas the highest level of accuracy was provided by transfer learning MobileNetV2 model with 89.04%. It was established that increasing depth led to better results, whereas Adam performed better optimizer compared to SGD regarding convergence.

Issues faced included overfitting, high computational resources, and lack of data. Possible improvements could involve collecting more data, implementing sophisticated data augmentation algorithms, and tuning MobileNetV2 architecture further.